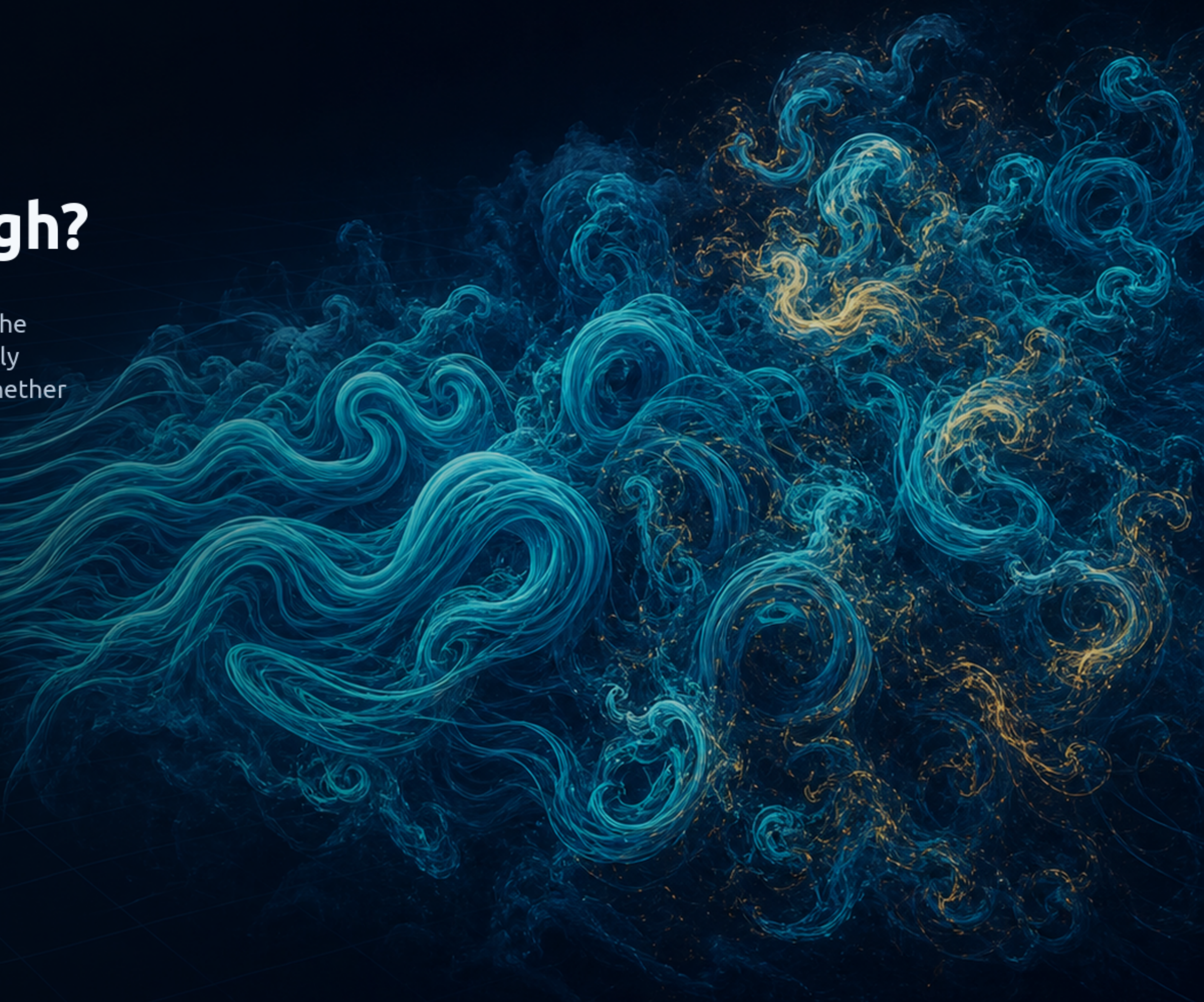
NAVIER-STOKES BRIEF

Can a smooth fluid become infinitely rough?

The 3D Navier–Stokes equations describe water and air. The Millennium Problem asks whether every smooth, physically reasonable starting flow stays smooth for all time—or whether a singularity can form in finite time.



THE QUESTION IN ONE LINE

Smooth forever or **finite blow-up?**

No generally accepted proof decides this in three dimensions.

Our repository investigates one proof route computationally.

It has not solved the Millennium Problem.

Conceptual illustration — not a simulation result

Stretching creates small scales; viscosity removes them

The unresolved issue is whether the smoothing always wins before any derivative becomes unbounded.

$$\partial u/\partial t + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0$$

velocity changes = transport + pressure + viscous smoothing; incompressibility forbids sources or sinks

VORTEX STRETCHING

Can sharpen gradients

- Three-dimensional vortices can stretch and intensify.
- Energy may move toward smaller spatial scales.
- A proof must rule out uncontrolled concentration.

NONLINEAR TERM $(u \cdot \nabla)u$

UNKNOWN BALANCE

VISCOSITY

Smooths the flow

- Diffusion damps high-frequency motion.
- Energy decreases in the classical energy estimate.
- Known estimates are not yet strong enough to close 3D regularity.

DIFFUSIVE TERM $\nu\Delta u$

A successful proof must control every scale uniformly—not only every scale tested by a computer.

NOT SOLVED

Human-directed, AI-executed falsification—not an automated proof

Codex wrote and ran the C++ engine, searched for hostile states, read failed routes and refined the next obligation while preserving restartable evidence.

LAYER	WHAT THE PROJECT DOES	CURRENT RESULT	STATUS
Spectral engine	Finite Fourier / Galerkin Navier–Stokes algebra	Reproducible C++20 kernels and certificates	VERIFIED
Gradient engine	JVP/VJP, RK4 adjoint checks and L-BFGS search	Fast adversarial state optimization	VERIFIED
Far-tail analysis	Separate sufficiently distant frequency shells	Cutoff-independent partial lemma	PROVED
PNT-13	Test standalone shell decorrelation	Near-perfect gap correlations found	FALSIFIED
PNT-12	Keep tail mass, Gram row and normalization coupled	Finite record 4.18215e-4; no theorem	OPEN

WHY THIS CAN SHORTEN RESEARCH

Codex can attack a weak lemma in seconds or minutes, store its counterexample, read the failure and redesign the next candidate around what survived.

WHAT REMAINS

Prove PNT-12, close PNT-4 and RQ-11, then complete the L4 to L5 to L6 chain.